SMART TRAFFIC MANAGEMENT SYSTEM USING AI

Capstone-I project report submitted

by the student of

Hybrid UG program in Computer Science & Data Analytics

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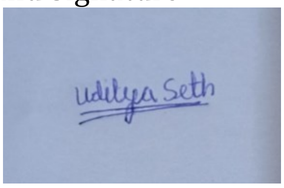
Date..................**20/04/2024**

Declaration

I hereby declare that this submission is my own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Date: Student Name, Roll No, Group No., and Signature

20/04/2024 **Uditya Seth, 2312res962, 94**



Summary of the Project

This project is based on Smart Traffic Management System Using Ai in this project we do research and do implementation and our aim is to aquired knowledge and firstly we have discuss and start working on object detection We do research on different platform and I have done short term course on object detection in which I have learn basic knowledge of object detection and Then I have applied codes of that I have learn but it was not working and after that during learning I get to know that there are two methods of object detection in python in yolo library & open cv so the course I have done in that I have learn yolo library but the codes I have applied it was not working then I choose 2 method of open cv then I research from different platforms than I found one yt and I try there codes and it was succesfull and then I work on video implementation from code and I have research it and learn it after then we start modifying our code and and we are trying to add speed detection in it and we think to take 5 or 4 videos from different source and name them cam1 to 5 and we do implement object detection on them and we think to add new features like speed detection and we add limitation of 40 km/hr if speed of that object is greater than than it will take a snapshot of that object and we create a link for our source video and our future plan is to build a source website and our source link will be there and the camera take the snap shot will save on our website and then we try to improvise our website by adding new features in it that snap shot it save and it scan the snap shot and through ai server it find the person or car details so that he gets fines so our aim of the project is that person do not break the rules and avoid over speeding from our project we save many life also but due to the mentality of person in india it is difficult so we also think that if person breaks the rule he get fine if again he repeats the same thing so his fine gets doubled from next time because we try that the person will already fined his detail will save on website that is the idea we think for implementing our capstone project we are new it may be possible that we can do every thing or may be half of it because our project is so long that it need times of max 1 years we try our best and I hope u see our efforts I end my summary with regards.

Contents

Chapter 1

Introduction

As we know that our population is increasing day by day and we know that in every home each family member have there personal vehicle so it cause many traffic problem and our old traffic management system cannot Handel these problem so for that problem we bring smart solution is SMART TRAFFIC MANAGEMNT SYSTEM USING AI our project is software based and our moto of this project is to make the optimize the flow of traffic without any natural loss

Advantages

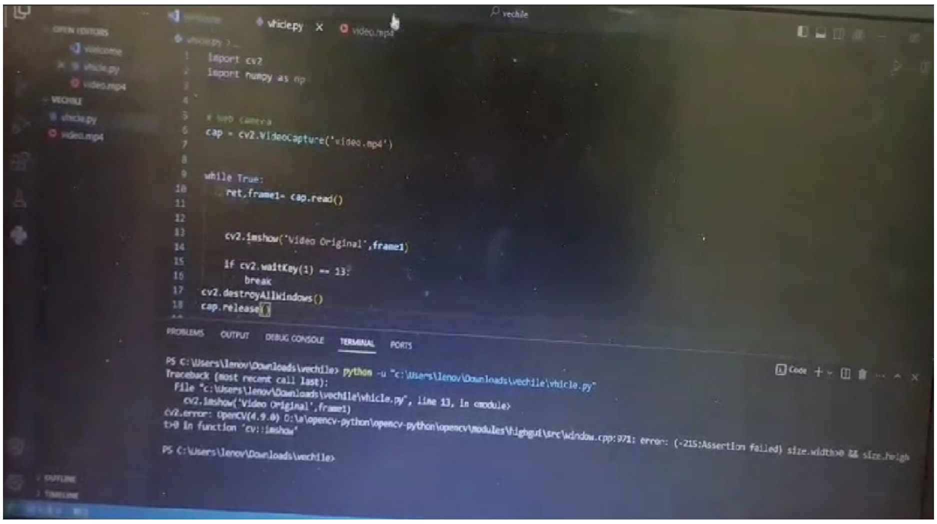
1> Our system is avoiding traffic jam
2> Our system avoids accidents
3> Our system avoids over speeding
4> Our system help to learn people not break the rule
5> Our system help to optimize the flow of vechicle
6> Our system saves our time

In this way a new technology (AI) help us to manage the system of traffic in our project we have use 4 or 5 videos and we do implementation of object detection and speed detection after that we use AI to make or work easy for doing traffic break fine so that how it helps to optimize the flow of traffic

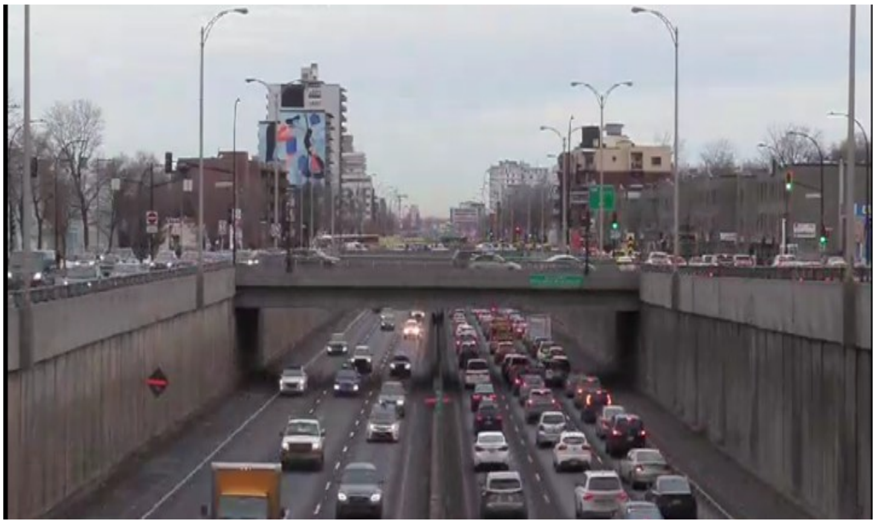
Chapter 2

Object Detection & Video Import

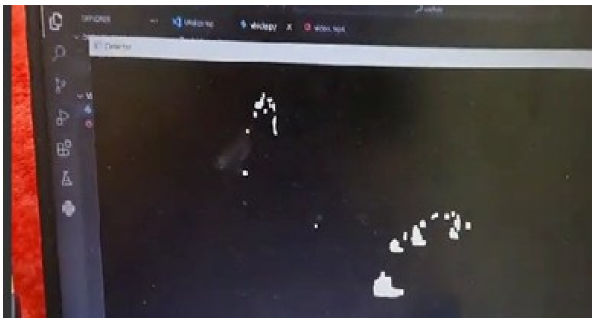
So as the project team descussion we all together decide to work on object detection before that work I start do research on video import after watching 15 video and I found one of the best video which was succesfully applied after completion of my task now I want some knowldege regarding object detection so I seaarch the history of object detection and after words I found a short paid certified course of platform cvzone in which the instructor give some theory knowledge on object detection and from that video I get to know there are two method of object detection from python 1> yolo library 2> open cv the course I was doing it gives knowledge of yolo library after completion of that course I start implementeing of codes of that teacher taught in course but codes were not working it after that I think to try second method of open cv and I started searching regarding open cv after words I found one channel on youtube and the code he tell I try to apply and it was succesfully done and after that the things I have learnt I try to implement it so first took 4 videos randomly from gooogle and firstly try to import it and after 3 attempt I have succesfully upload it on vs code and I try the code which I have learn in that apply only in first video in which I the background was black and object come is white after that I try it and it was succesfully worked after than I worked on code of object detection so I tried the codes at the result there was find anything moves in video it dtects the object after words I try the codes on that video know I counting of my object and now I try to search from other sources I have found one source and I apply the codes it work succesfully now on next page I want to show some pics and codes of my through which I can elebrate my work

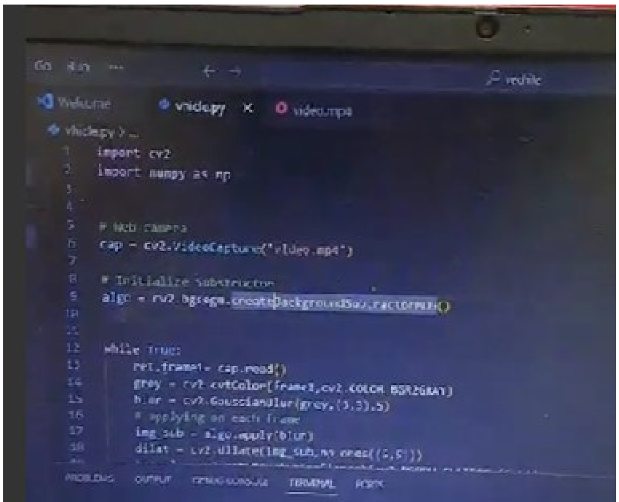


So the pic above is the pic of code of importing video on vs which I have implemented in it and now in pic to I show the output of my video

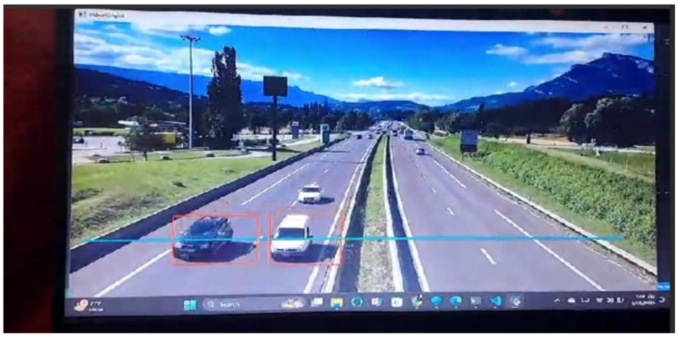


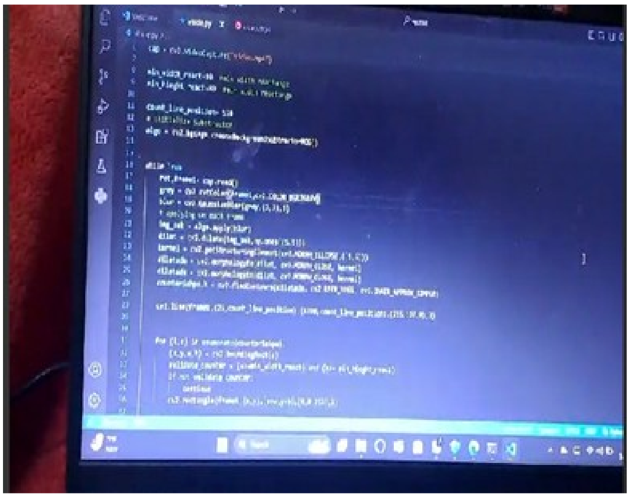
Now I going to show my second part of work in which I have done my background video black and object is white with codes



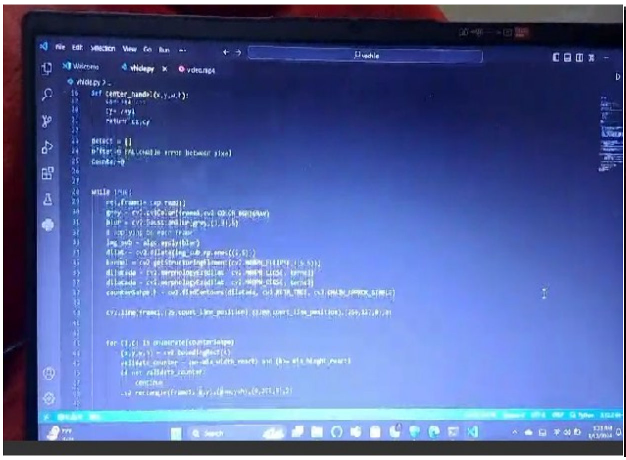


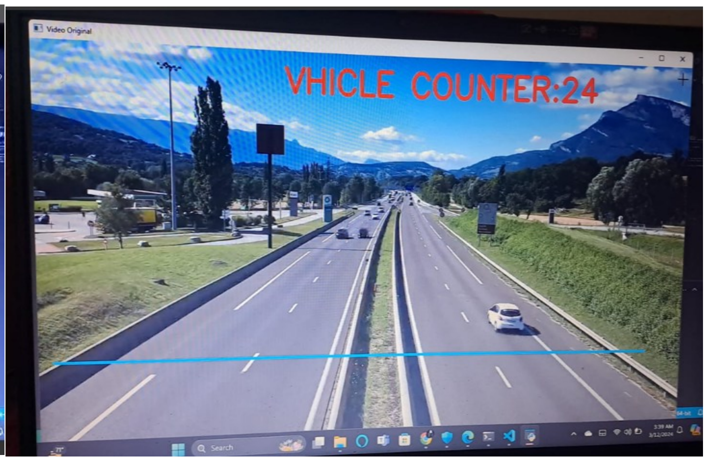
After I want to show the object detection code with output





Now I want to show my code of vhicle count ……….





Code explination

1> Video importing

Now I would like to explain my code my first code is video import so in my first have import open cv and then I have applied numpy after that I use the video import syntax and after that I have apply while loop statement and take a refrence of frame in which we need our video after that I use if statement for the loop after that I use break for terminate the loop after than I write ending syntax after than we run the code video which I want to input for further work it runs successfully the code we use for any kind of video importing after I move furthur procees and I modified my code in which I want to make background of my

2 > Background color code

Now I have try to modified the code and I want to change the background colur of my code so the code is same till video import syntax after than I initialize the substructor algorithm syntax for changing background colour then till frame of refrence code is same after the next line we the colour code and we give some colour numbers after then we write the colour change code with respect to ftame of that video after then I use some k

Then I use some kernels syntax and after the rest code will remain same and then after successfully applied that algorithm code we are able to change background of my video

3> Object Detection code

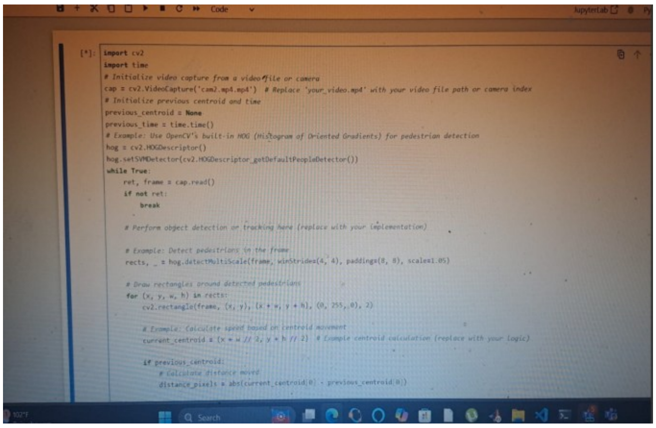
After the background change code then i have modified the code again and first I try to apply the line in code so that when object cross the line it easily to take snapshot of that vhicle after we use object detection algorithm in our code and it works successfully after than I start to modified my code for vhicle count it also works succefully after then I try to find the codes of speed detection but I found many channels I try to learn the code but the problem occurs everything was correct but the code cant run on my system then I use the other sources for the code but the problem is same and some times the code   gets terminates so at the end I took some help of my btech seniors of hostel then the problem that occur in my code they help me a lot for speed detection problem after than my code runs successfully but the time came for submission so I have done that much work till my mid sem report

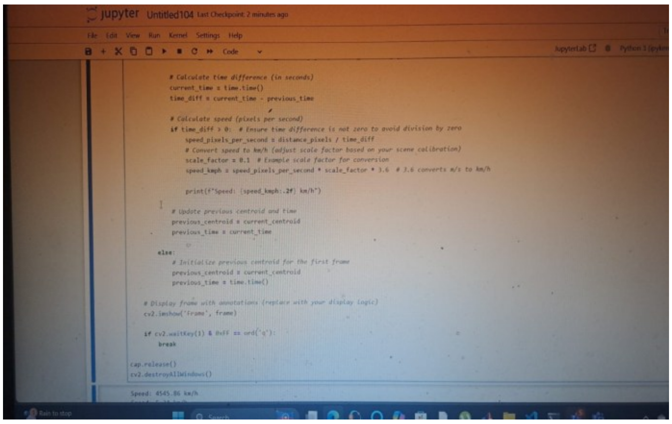
SPEED DETECTION

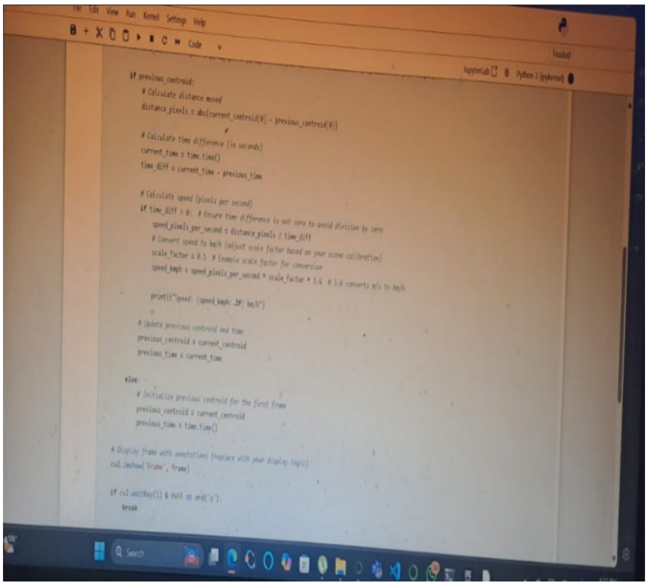
Abstract – Road accident has been very common in present world due to careless driving with the help of new technology we can control this problem of over speed driving at this scenario we are praposing a system which detect the speed of vhicle our over all project is devided into different parts object detection, speed detection, vhicle counter , number plate detection, helmate detection And I have done work on object detection and speed detection.

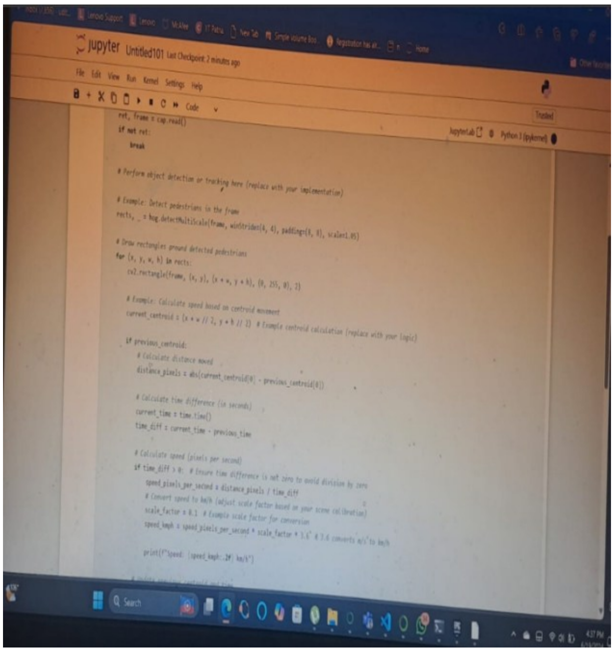
After completion of mid sem work we all arrange a meeting in which we decided what next work we have to do in project after discussion I have started my work on speed detection first I need to gain knowledge regarding some basic concept of speed detection for that I have search some course but I did get it then I read some articles and see some videos after that I get to know that I can do speed detection by two methods 1> open cv 2> yolo library so I choose yolo library first for that I search that what package is required to install for yolo library then I started install yolo library

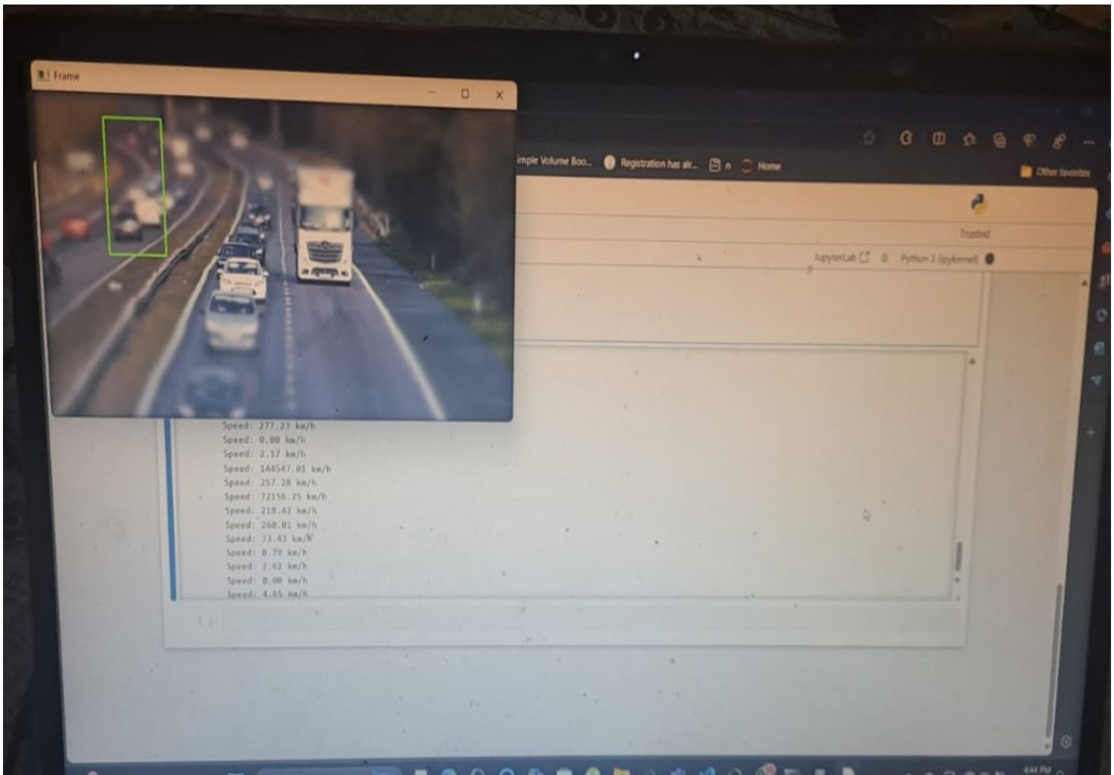
CODE AND OUTPUT











CODE EXPLINATION

We use import cv2 and import time these are library which are use in speed detection after we use cap = cv2.VideoCapture('video path') – this syntax is use to initialize the video from our pc storage and then we use previous_centroid = none & previous_time.time() these both syntax is use to initialized previous centroid and previous time i use the syntax hog to break in this syntax I use built in function hog histogram of oriented gradients for strain detections then I use frame of video then we applied some calculation on frame and give them coordinates after that I use if statement in time difference speed of pixels of the video and I apply the formula of pixels per km/hr

after else statement I use syntax to go for previous centroid and previous time at last cap release and all windows destroy.

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Conclusion & future work

Our conclusion of the project is that we develop new smart technology through which we can optimize the flow of traffic and we aim that from our technology we can reduce natural losses and if in future we try to improve and advanced this system then it may be a big change for our future generation and there are many things that me and my teametes has done like object detection ,speed detection ,number plate detection ,vhicle counter and helmate detection but we were left some part to develop in this project so we decided our future work that in future we develop website and with the help of ai we can find the identity of vehicle and info regarding the person through screen short so it may be easy to get is rule fine and the collected data will remain save on our website if the person repeats the same braked the rule then our ai search info that person is already fined or new if person who was fine earier and if he was the same person then the fine gets 3 times of the previous one and we hope from our system we can save many life and and we try to reduced the traffic jams and from this system we try to optimize the flow of traffic we make 80 % project but we left 20% and we try our best to do the as much as we can because of online connectivity and teamwork it is not possible to complete it but from my side I try to give 100% for this project hope u can understand it.

References

1> The object detection course on yolo lib – cv zone

2> Open cv library on object detection - code with kiran

3> Chat gpt for theory knowledge of object detection

4> Google - some basic knowledge of speed detection

5> Github - some code knowledge

6> Seniors help - for some errors in the code

7> Pw skills – for learning basics of python

8> Code with aarohi – for speed detection in yolo library

9> Pyresearch – for speed detection in open cv

10> For some code error finding I use copilete

11> Pixels vide0

12> Hickvision

13> Articles